A Compact c_0 Subset With No Lipschitz p-Approximate Schauder Frame

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Status

This packet gives a counterexample to the universal part of Problem 3.1(1) in Krishna–Johnson, arXiv:2211.10652 [1]. Not every subset of a Banach space has a Lipschitz p-approximate Schauder frame for some $1 \leq p < \infty$. In fact, there is a compact subset of a Banach space isometric to c_0 which has no Lipschitz p-ASF for any finite p.

Source Problem

The source paper formulates the following open problem on PDF page 10.

> Motivated from the approximation properties of Banach spaces (Schauder basis problem) [5, 11] and from the failure of atomic decompositions for (even separable) Banach spaces (see [6]), we formulate the following interesting (high-end) problem.
>
> **Problem 3.1.** *(1) Classify which subsets of a Banach space has Lipschitz p-ASF for some $1 \leq p < \infty$? In particular, does every subset of a Banach space has Lipschitz p-ASF for some $1 \leq p < \infty$?*
>
> *(2) Classify which subsets $\mathcal{M}$ of a Banach space $\mathcal{X}$ has the following property. There exists a sequence $\{\tau_n\}_n$ in $\mathcal{M}$ and a sequence $\{f_n\}_n$ in $\operatorname{Lip}(\mathcal{M}, \mathcal{X})$ such that*
>
> $$x = \sum_{n=1}^{\infty} f_n(x)\tau_n, \quad \forall x \in \mathcal{M}.$$
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Figure 1: Problem 3.1 from arXiv:2211.10652, PDF page 10.

The construction below answers the "in particular" question in item (1) negatively. It does not claim a classification of all subsets and does not settle item (2)'s weaker representation problem.

Definitions Used

For a subset $\mathcal{M} \subseteq \mathcal{X}$, a Lipschitz p-ASF in the sense of [1] consists of scalar coefficient maps f_n, points $\tau_n \in \mathcal{M}$, an analysis map

$$\theta_f : \mathcal{M} \to \ell^p, \qquad \theta_f(x) = (f_n(x))_{n=1}^{\infty},$$

a bounded synthesis operator

$$\theta_\tau : \ell^p \to \mathcal{X}, \qquad \theta_\tau((a_n)) = \sum_{n=1}^{\infty} a_n \tau_n,$$

and a Lipschitz frame map

$$S_{f,\tau} = \theta_\tau \theta_f : \mathcal{M} \to \mathcal{M}$$

which is invertible and bi-Lipschitz.

Proof Intuition

The analysis map of a Lipschitz p-ASF is not merely a coefficient map: it must bi-Lipschitzly encode the metric of $\mathcal{M}$ into ℓ^p. Indeed, the frame map is the synthesis operator applied to the analysis map, and the frame map has a Lipschitz inverse. Thus a subset with no bi-Lipschitz embedding into any ℓ^p, $p < \infty$, cannot have a Lipschitz p-ASF.

The counterexample is a compact bouquet in c_0 containing smaller and smaller copies of the balls of ℓ_∞^n for all n. Compactness comes from shrinking the n-dimensional pieces to zero. The obstruction is local: a bi-Lipschitz embedding into a fixed ℓ^p would, on every finite-dimensional piece, produce uniformly distorted embeddings of ℓ_∞^n into ℓ^p. Rademacher differentiation turns these into linear embeddings, while cotype of ℓ^p forces the distortion to grow with n.

Lemma 1. *If $\mathcal{M} \subseteq \mathcal{X}$ has a Lipschitz p-ASF for some $1 \le p < \infty$, then $\mathcal{M}$ admits a bi-Lipschitz embedding into ℓ^p.*

Proof. Let $A = \theta_f : \mathcal{M} \to \ell^p$, $V = \theta_\tau : \ell^p \to \mathcal{X}$, and $S = VA : \mathcal{M} \to \mathcal{M}$. By definition, A is Lipschitz, V is bounded, and S is bi-Lipschitz. Thus there are constants $L, m > 0$ such that

$$\|Ax - Ay\|_p \le L\|x - y\|, \qquad \|Sx - Sy\| \ge m\|x - y\|$$

for all $x, y \in \mathcal{M}$. Since $S = VA$,

$$\|Ax - Ay\|_p \ge \frac{1}{\|V\|}\|V(Ax - Ay)\| = \frac{1}{\|V\|}\|Sx - Sy\| \ge \frac{m}{\|V\|}\|x - y\|.$$

Hence A is a bi-Lipschitz embedding of $\mathcal{M}$ into ℓ^p. □

Lemma 2. *There is a compact set $\mathcal{M} \subset c_0$ which does not admit a bi-Lipschitz embedding into ℓ^p for any $1 \le p < \infty$.*

Proof. Let $E_n = \ell_\infty^n$ and let

$$X_0 = \left(\bigoplus_{n=1}^{\infty} E_n \right)_{c_0}.$$

This Banach space is isometric to c_0, after enumerating the finitely many coordinates in each block. Put $r_n = 2^{-n}$. We regard E_n as the n-th coordinate block of X_0, and define

$$\mathcal{M} = \{0\} \cup \bigcup_{n=1}^{\infty} r_n B_{E_n},$$

where $r_n B_{E_n}$ is placed in the n-th block and all other blocks are zero.

The set $\mathcal{M}$ is compact. Indeed, for every $\varepsilon > 0$ all but finitely many blocks lie in the ε-ball around 0, and the remaining union is finite-dimensional compact.

Suppose, towards a contradiction, that $F : \mathcal{M} \to \ell^p$ is bi-Lipschitz for some $1 \le p < \infty$. Then there are constants $0 < a \le b < \infty$ such that

$$a\|x - y\|_{X_0} \le \|F(x) - F(y)\|_p \le b\|x - y\|_{X_0} \qquad (x, y \in \mathcal{M}).$$

Fix n. On the relative interior of the block $r_n B_{E_n}$, define

$$\Phi_n(u) = r_n^{-1} F(r_n u), \qquad u \in (-1, 1)^n \subset \ell_\infty^n.$$

Then

$$a\|u - v\|_\infty \le \|\Phi_n(u) - \Phi_n(v)\|_p \le b\|u - v\|_\infty \qquad (u, v \in (-1, 1)^n).$$

Since ℓ^p has the Radon–Nikodym property for $1 \le p < \infty$, the Banach-valued Rademacher theorem gives a point of Fréchet differentiability of Φ_n. Let

$$T_n : \ell_\infty^n \to \ell^p$$

be the derivative at such a point. Passing to difference quotients in the upper and lower Lipschitz inequalities yields

$$a\|h\|_\infty \le \|T_n h\|_p \le b\|h\|_\infty \qquad (h \in \ell_\infty^n).$$

Now let $q = \max\{p, 2\}$. The space ℓ^p has finite cotype q; write C_p for a cotype q constant. Applying the cotype inequality in ℓ^p to the vectors $T_n e_1, \ldots, T_n e_n$, where e_i is the canonical basis of ℓ_∞^n, gives

$$\left(\sum_{i=1}^{n} \|T_n e_i\|_p^q \right)^{1/q} \le C_p \, \mathbb{E}_\varepsilon \left\| \sum_{i=1}^{n} \varepsilon_i T_n e_i \right\|_p .$$

The lower estimate on T_n gives the left side at least $a n^{1/q}$. The upper estimate gives the right side at most

$$C_p b \, \mathbb{E}_\varepsilon \left\| \sum_{i=1}^{n} \varepsilon_i e_i \right\|_\infty = C_p b.$$

Thus

$$\frac{b}{a} \ge \frac{n^{1/q}}{C_p}$$

for every n, an impossibility. Therefore $\mathcal{M}$ cannot bi-Lipschitz embed into any ℓ^p, $1 \le p < \infty$. $\square$

Theorem 1. *There is a compact subset $\mathcal{M}$ of a Banach space isometric to c_0 such that $\mathcal{M}$ has no Lipschitz p-approximate Schauder frame for any $1 \le p < \infty$.*

Proof. Take the compact set $\mathcal{M} \subset c_0$ from Lemma 2. If $\mathcal{M}$ had a Lipschitz p-ASF for some $1 \le p < \infty$, then Lemma 1 would give a bi-Lipschitz embedding of $\mathcal{M}$ into ℓ^p, contradicting Lemma 2. $\square$

Verification Notes

The only external geometry used is standard: ℓ^p has cotype $\max\{p, 2\}$, and Lipschitz maps from finite-dimensional domains into Banach spaces with the Radon–Nikodym property are differentiable almost everywhere. The proof does not require finite-dimensional pieces to be linearly complemented in the target; differentiating a hypothetical bi-Lipschitz embedding supplies the linear maps T_n.

The compactness check is also important. The pieces $r_n B_{\ell^n_\infty}$ are infinite compact sets, not finite nets, but their radii tend to zero. Thus every sequence either has infinitely many terms in a finite union of compact blocks or eventually lies arbitrarily close to 0.

Novelty and Search Bounds

On June 16, 2026, the run indexes were searched for arXiv:2211.10652, "Lipschitz p-Approximate Schauder Frames", "Lipschitz p-ASF", "approximate Schauder frame", and variants. No prior packet or attempt for this source was found.

External web/arXiv searches for the exact title, "Lipschitz p-ASF every subset", "Classify which subsets Lipschitz p-ASF", and "Lipschitz approximate Schauder frame subset Banach" found the source paper and adjacent frame papers by the same authors, but no later solution to Problem 3.1 and no counterexample of the compact c_0 form above. The packet should therefore be reviewed as a new counterexample to the universal part of Problem 3.1(1), subject to human verification.

Limitations

This does not classify all subsets admitting Lipschitz p-ASFs. It also does not settle Problem 3.1(2), whose stated representation property is weaker than the Lipschitz p-ASF structure used in Lemma 1.

Human Review Recommendation

Review as a counterexample packet. The main points to check are:

1. the implication from Lipschitz p-ASF to bi-Lipschitz embeddability into ℓ^p;
2. the compactness of the block bouquet $\mathcal{M} \subset c_0$;
3. the differentiation/cotype obstruction for uniformly embedding all ℓ^n_∞ balls into one ℓ^p.

References

[1] K. Mahesh Krishna and P. Sam Johnson, *Lipschitz p-Approximate Schauder Frames*, arXiv:2211.10652.

[2] Y. Benyamini and J. Lindenstrauss, *Geometric Nonlinear Functional Analysis, Volume 1*, American Mathematical Society Colloquium Publications, vol. 48, 2000.

[3] M. Ledoux and M. Talagrand, *Probability in Banach Spaces*, Springer, 1991.